Power Supplies

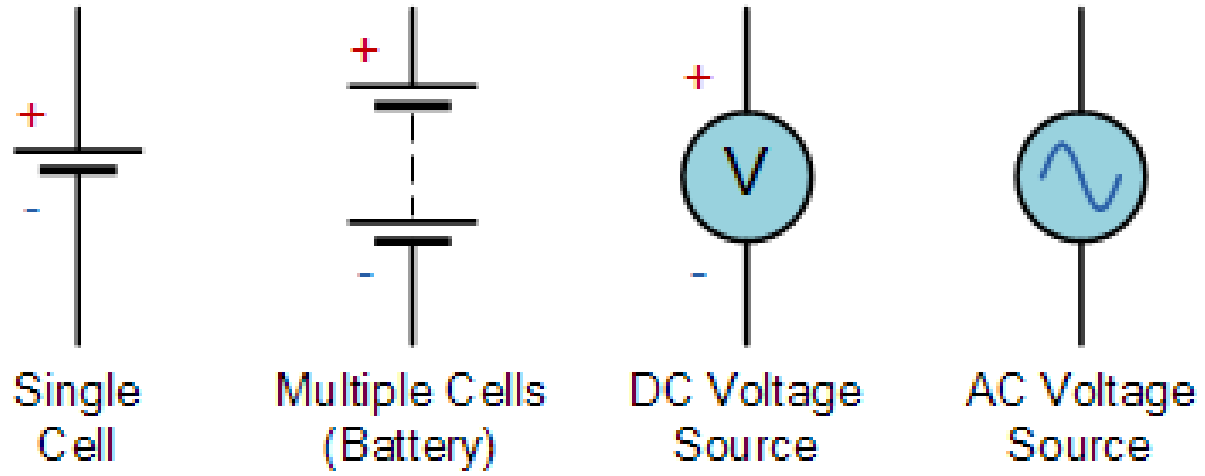
Voltage

Voltage is the driving force in an electrical circuit that pushes electrons through a conductor, creating an electric current that can perform work, such as lighting a bulb, powering a motor, or charging a device. Think of voltage as the "pressure" that motivates the flow of electrons, similar to how water pressure pushes water through pipes.

Voltage is measured in **volts (V)**, named after the Italian physicist **Alessandro Volta (1745–1827)**, who invented the **voltaic pile**, the first chemical battery capable of producing a steady electric current. This invention laid the foundation for modern electrical circuits.

Historically, voltage was referred to as **electromotive force (emf)** because it "drives" electrons along a circuit, much like a force moves objects in mechanics. In many fundamental equations of electricity, including **Ohm's Law**, voltage is commonly represented by the symbol **E**, reflecting this older terminology.

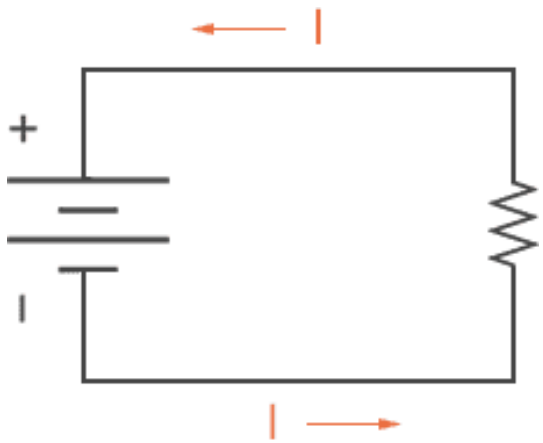
VOLTAGE SYMBOLS



Voltage is either **Alternating Current (AC) Voltage** or **Direct Current (DC) Voltage**. Ways they differ:

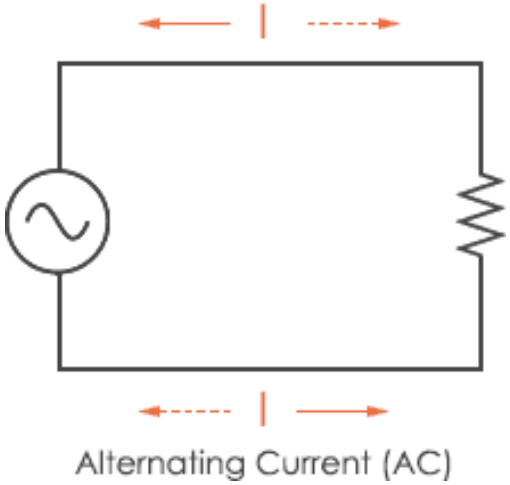
Direct Current (DC)

Direct current (DC) is the type of electrical current in which electrons flow **in a single, constant direction** through a conductor. Because the flow is steady, DC is ideal for powering devices that require a consistent voltage, such as **batteries, flashlights, mobile phones, and most electronic gadgets**.



Alternating Current (AC)

Alternating current (AC) is a type of electrical current in which the flow of electrons **periodically reverses direction**. This means the voltage also alternates between positive and negative. AC is the standard type of electricity supplied to homes and businesses because it can **travel long distances efficiently,** and its voltage can be easily increased or decreased using transformers.



Batteries and Their Types

Batteries are devices that **store chemical energy and convert it into electrical energy (DC)**. They provide power to circuits when no external power source is available.

Common types of batteries:

- **Primary Batteries:** These are **non-rechargeable**, like **AA, AAA, or 9V batteries** used in remote controls and flashlights.
- **Secondary Batteries:** These are **rechargeable**, like **lithium-ion (Li-ion)** and **nickel-cadmium (Ni-Cd)** batteries found in phones, laptops, and cameras.

Note: Batteries are vital for portable electronics because they supply DC voltage directly to circuits.

Rectifiers, Regulators, and Filters

When we use AC power to run DC devices, we need a **power supply circuit** that converts AC into DC. This process involves three key components:

1. **Rectifier** – Converts AC into pulsating DC.
    - **Example:** A **diode bridge rectifier** changes AC from the wall outlet into DC.
2. **Filter** – Smooths the pulsating DC by removing ripples using **capacitors**.
    - The result is a steadier DC output.
3. **Voltage Regulator** – Keeps the DC voltage **constant and stable**, even if the input voltage changes.

- **Example:** The **7805-voltage regulator** provides a constant 5V output.

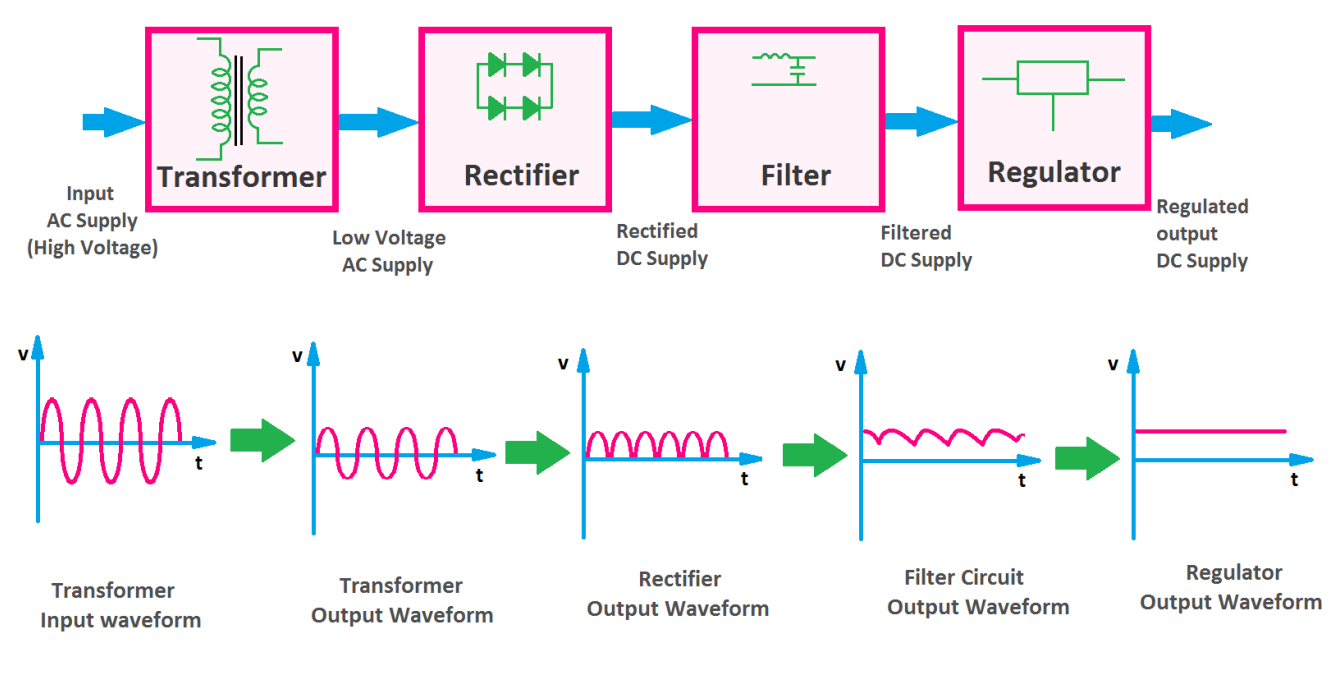
Together, these components form the core of most **DC power supplies**
used in electronic circuits.

Introduction to Power Supply Design

A **power supply** is a circuit that provides the correct voltage and current to operate electronic devices. It converts AC from the wall outlet into a stable DC voltage.

A **basic power supply design** includes:

1. **Transformer** – Reduces high AC voltage to a lower level.
2. **Rectifier** – Converts AC to DC.
3. **Filter** – Smooths the DC signal.
4. **Regulator** – Maintains a steady DC voltage.



Reference:

https://byjus.com/physics/difference-between-ac-and-dc/